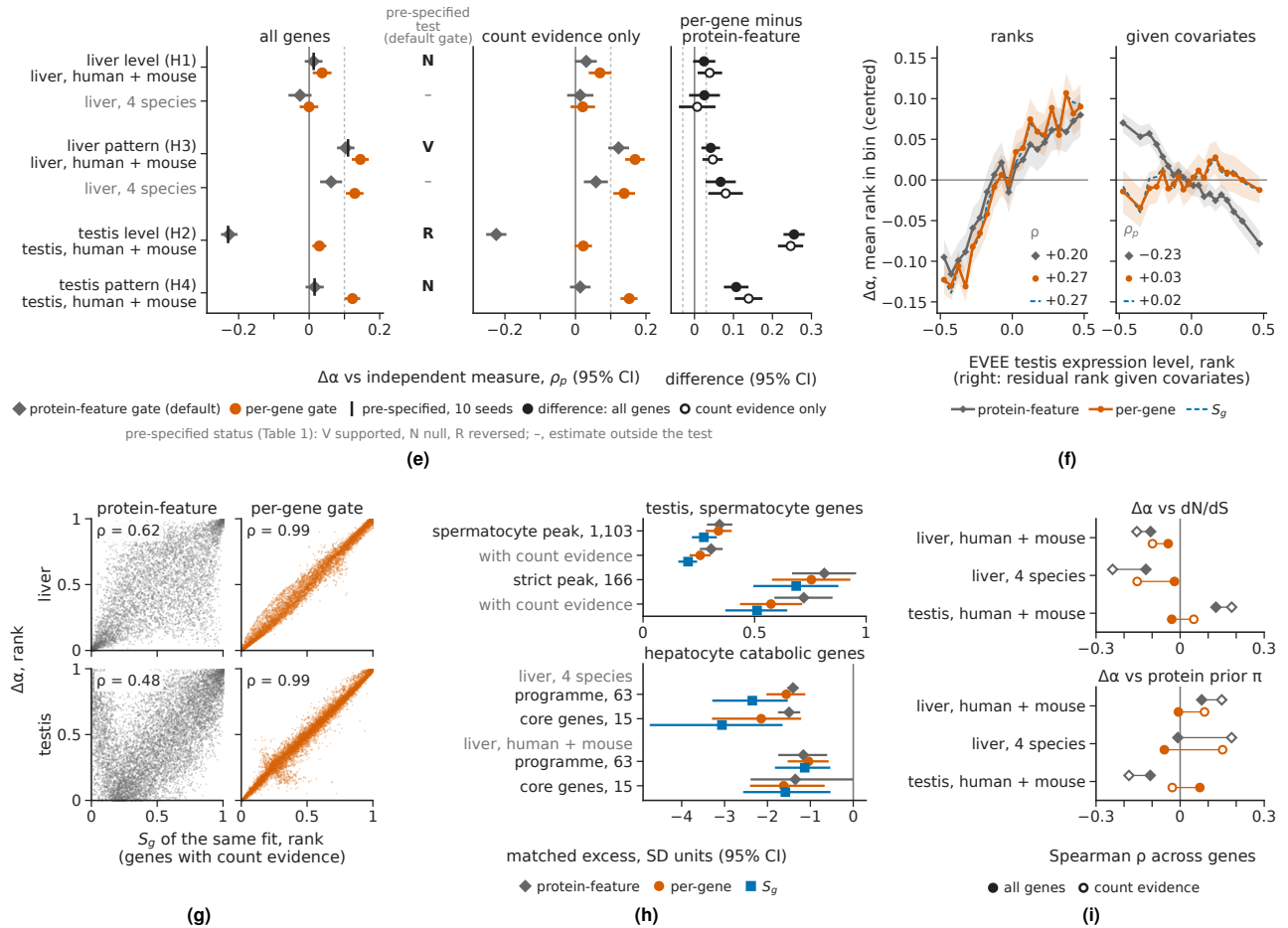

all genes
pre-specified test (default gate)
count evidence only
per-gene minus protein-feature
liver level (H1)
liver, human + mouse
liver, 4 species
liver pattern (H3)
liver, human + mouse
liver, 4 species
testis level (H2)
testis, human + mouse
testis pattern (H4)
testis, human + mouse
N
–
V
–
R
N
−0.2
0
0.2
0
0.1
0.2
0.3
Δα vs independent measure, ρ_p (95% CI)
difference (95% CI)
protein-feature gate (default)
per-gene gate
pre-specified, 10 seeds
difference: all genes
count evidence only
pre-specified status (Table 1): V supported, N null, R reversed; –, estimate outside the test
(e)
ranks
given covariates
Δα, mean rank in bin (centred)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
ρ
+0.20
+0.27
+0.27
ρ_p
−0.23
+0.03
+0.02
−0.5
0.0
0.5
EVEE testis expression level, rank
(right: residual rank given covariates)
protein-feature
per-gene
S_g
(f)
protein-feature
per-gene gate
ρ = 0.62
ρ = 0.99
ρ = 0.48
ρ = 0.99
liver
testis
Δα, rank
0
0.5
1
S_g of the same fit, rank
(genes with count evidence)
testis, spermatocyte genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
hepatocyte catabolic genes
liver, 4 species
programme, 63
core genes, 15
liver, human + mouse
programme, 63
core genes, 15
−4
−3
−2
−1
0
matched excess, SD units (95% CI)
protein-feature
per-gene
S_g
(h)
Δα vs dN/dS
liver, human + mouse
liver, 4 species
testis, human + mouse
−0.3
0
0.3
Δα vs protein prior π
liver, human + mouse
liver, 4 species
testis, human + mouse
Spearman ρ across genes
all genes
count evidence
(i)
(g)